Instructions Buy the Phrase; the Phrase Costs Preference: an Epistemic Audit of LLM Aggregation Pipelines at Auditable Scale

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Abstract

Mixture-of-Agents (MoA) pipelines — proposer models whose outputs an aggregator synthesizes — achieve state-of-the-art results on preference-judged benchmarks, but every published gain is measured by an LLM preference judge, and nothing in that literature measures what such gains are made of. We audit the MoA architecture at auditable scale (7–20B local models) with pre-registered cells, validated instruments, and causal designs, and report three linked findings. (1) A phrase-swap experiment shows that an epistemic instruction ("label estimates with phrase X") produces *only* its named phrase: the dictated form appears in two thirds of outputs regardless of which synonym is named, while the construct rate outside the named phrase is statistically unchanged (CIs [−0.079, +0.111], [−0.103, +0.087]). (2) Order-debiased preference judging shows a real aggregation lift — the aggregated pipeline beats the same writer answering directly in 0.818 [0.718, 0.919] of decisive pairs, surviving length matching and replicating under a second judge from a disjoint model family (0.788 [0.645, 0.917]) — yet the same judges *penalize* the instructed arms: the phrase-bearing arms lose to their own control (0.261 [0.140, 0.388]; 0.317 [0.152, 0.480]; the penalty holds in three of four judge×form cells, with the fourth agreeing in point estimate at lower power). Because the arms differ only in the phrase, the penalty is causally attributable to it. Together these results identify a measured tension: instruction tuning of epistemic disposition installs surface forms, and preference optimization removes exactly those forms. (3) The preference instrument itself is majority reading-order at this scale, as a property of the task format: raw first-position preference is 0.85–0.88 across four judges from four model families, and order-debiasing converts 60–95% of pairs to ties. We further report mechanism results that bound the MoA narrative — the aggregator selects sources rather than recomputing, ignores cross-proposer corroboration, and transports only a small fraction of the epistemic qualification it is given, with within-corpus slopes of 0.11–0.33 replicated across two writers and two corpora. The audit's design consequences were then engineered and prospectively re-tested as their own registered cells: assigning a contested quantity to a second proposer halves corrupted-value adoption (0.90 → 0.45); carrying epistemic qualification out-of-band, in a mechanically assembled appendix the writer never sees, preserves all of it against the 17% that survives prose, measurably changes downstream decisions (+0.69 over a matched irrelevant-appendix control, at parity with in-prose placement), and shows no detected preference cost; and the re-consultation loop closes end-to-end with a *live* proposer (adoption 0.905 vs. 0.333, within noise of a scripted ceiling). We release the measurement kit, the dictation registry, and all pre-registrations.

1 Introduction

Mixture-of-Agents [1] reports that a stack of open-source proposers plus an aggregator surpasses GPT-4o on AlpacaEval 2.0 (65.1% vs. 57.5% length-controlled win rate), and frames the result as "collaborativeness": models produce better responses when shown other models' outputs, even lower-quality ones. Every number in that paper — and in the surrounding multi-agent literature — is produced by an LLM preference judge. No published MoA evaluation measures a verifiable outcome, tests error propagation, validates its judge on the deployed task, or separates behavioral change from surface compliance.

This paper supplies that audit. It is the product of a falsification-first research program that ran more than fifty pre-registered experiment cells on a MoA-architecture pipeline small enough to audit completely: 7–20B open models on consumer hardware, with every registration, deviation, and verdict recorded in a public ledger before and after each run. The architecture is a two-layer MoA with three proposers and one aggregator. Earlier program cells ran the MoA Aggregate-and-Synthesize prompt *verbatim*; the causal cells of §5–§6 deliberately use a minimal aggregator prompt instead, so that instruction content is the manipulated variable rather than a background condition.

Three questions organize the audit:

1. **What do epistemic instructions to an aggregator actually do?** (§5) A counterbalanced phrase-swap cell shows the answer is: install the named phrase, and nothing measurable beyond it.

2. **Does the MoA preference lift replicate at auditable scale, and what is it made of?** (§6) It replicates — 0.818 of decisive pairs, surviving a length control — and its composition resists our full surface feature set. The same judge penalizes instructed epistemic marking, causally.

3. **Do the mechanisms the MoA narrative assumes exist?** (§7) Largely no: the aggregator prefers uncorrupted sources but does not recompute; corroboration among proposers does not increase transport; two same-base domain fine-tunes do not share a reasoning signature.

The audit's instruments are themselves audited (§4): fourteen recorded instrument-validity findings drive the methodology, including the discovery — made mechanically by a dictation registry — that an earlier judge prompt in our own program enumerated the same phrases the system prompts dictated to writers, entangling instrument with scaffold at the instrument layer.

Contributions. (1) The first causal separation of compliance from behavior in epistemic instruction-following (the phrase-swap design), with a two-link causal chain showing preference optimization and epistemic disposition are in tension. (2) A replication of the MoA preference lift at 20B scale under an order-debiased protocol, with a decomposition showing it is not length, domain-framework density, or hedging register. (3) A measured characterization of local pairwise preference judging as majority reading-order. (4) Mechanism results bounding the collaborativeness narrative, and one positive: orchestrator-routed re-consultation, now closed end-to-end with a live proposer, with evidence that both halves of the loop are recognition-limited rather than capability-limited. (5) A prospectively re-tested architecture distilled from the audit — engineered redundancy, out-of-band epistemic freight, evidence-bound seating — in which every component carries its own registered verdict, including two informative nulls that demoted our own design rationales (§8). (6) An open measurement kit (`gst`), a dictation registry with a zero measured over-attribution rate, and the complete pre-registration ledger.

2 Related work

Aggregation pipelines. MoA [1] is the direct object of this audit. [2] showed that aggregating repeated samples of the single best model ("Self-MoA") outperforms mixed MoA by 6.6% LC on AlpacaEval — converging, from the benchmark side, with our finding that mixing is not automatically diverse (§7). Ensemble alternatives (rankers, routers, fusers) are surveyed in [1]; our LLM-ranker-adjacent result is the source-selector finding. [6] report that a single strong-prompted agent can match multi-agent discussion, consistent with our council nulls.

Judge pathologies. Position bias and verbosity bias in LLM judges are documented by [3]; length correlations in preference optimization by [4]; sycophancy under preference pressure by [5]. Our contribution is quantitative and causal at the pipeline level: an 85% raw first-position rate for a 20B judge, and a counterbalanced demonstration that specific dictated epistemic phrases lose preference.

Instruction following. Our phrase-swap design is, to our knowledge, the first to test an epistemic instruction by swapping its named surface form under otherwise byte-identical prompts, isolating form-independent behavioral residue from compliance.

3 The pipeline and the ledger

The audited pipeline is a two-layer MoA: three proposer seats (prompted or domain-fine-tuned 7–8B models, or role-prompted samples of the writer) whose outputs are concatenated for a 20B mixture-of-experts writer (`gpt-oss:20b`) under either the MoA Aggregate-and-Synthesize prompt or a minimal "lead analyst" prompt. The case battery is eighteen strategic advisory scenarios spanning healthcare, legal, and finance content.

Every cell is registered before it runs: hypotheses, falsification conditions, attainability computations (the fraction of runs in which the outcome variable can fire, computed from existing data, for every registered prediction), and consequences. Verdicts land in a status ledger that distinguishes LIVE, PROVISIONAL, WITHDRAWN, and BLOCKED claims; three program-level audits re-classified earlier results, and nulls are cited only at the strength their audit class permits (informative, weak, uninformative, or degenerate). All numbers below are LIVE ledger entries unless marked otherwise.

4 Instruments, audited

Fourteen recorded instrument-validity findings shape the methodology. Four matter most here.

Entanglement (finding 8). If a prompt dictates the strings an instrument detects, measurement M confounds compliance C with behavior B: $M = C + B$, and a perfect detector still counts an echoed phrase, because the difference between C and B is causal, not textual. A pattern-level decomposition (PD-13) first exposed this: an instruction moved its dictated phrasings from 0/30 to 28/30 while non-dictated phrasings of the same construct moved 1/30 to 4/30 (difference-in-differences +0.83 [+0.67, +1.00]).

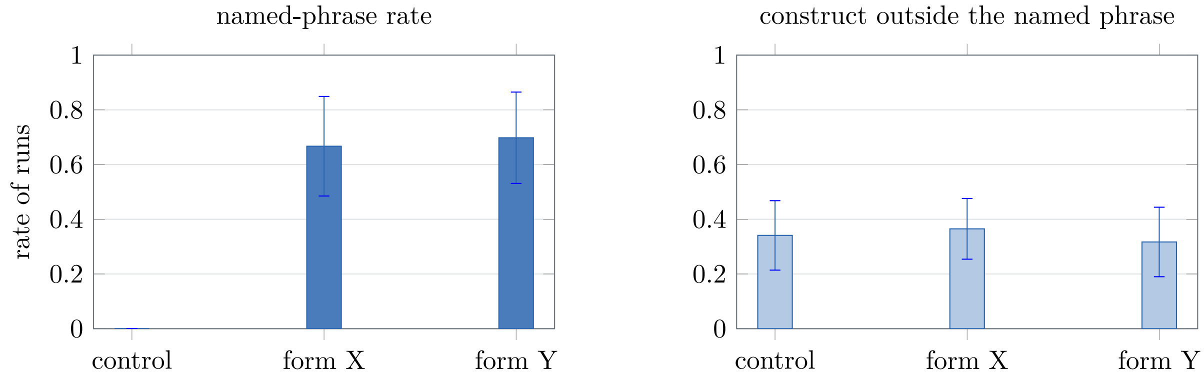

Figure 1: The phrase-swap cell (378 runs, cluster-bootstrap 95% CIs over 18 cases). **Left**: each instructed arm produces its own named phrase in about two thirds of runs; control never produces either, and cross-form leakage is 0/252. **Right**: the validated-judge construct rate *outside* the named phrase is statistically indistinguishable across all three arms. The instruction's entire measurable effect is the phrase it names.

The dictation registry. We extracted every construct-bearing phrase any system, generation, or judge prompt places in a model's mouth (125 entries from 35 prompt symbols; AST parsing and quote scanning, no semantic classifier), froze it under a hash, and gated all subsequent prompts against it. On its first run the registry found that an earlier pairwise judge prompt in our own program enumerated the same seven phrases the training-pair generators dictated to writers — instrument entanglement at the instrument layer (finding 12). Measured over-attribution is zero: across 2,907 clean-provenance spans (438,797 characters) and 60 descaffolded runs, no registry phrase appears without dictation upstream — direct probes confirm 0/60 for every signature phrase (finding 13). Registry surface is therefore *diagnostic of the scaffold*, which is what licenses literal-match partitioning even where a judged paraphrase stage is withdrawn.

The validated judge. Construct presence is scored by a two-judge, batched sentence-level protocol whose definitions name no phrase and instruct judges not to reward particular wording; it passes the registry gate, and agreement on the deployed corpus is 0.910 (14,088/15,485 decisions). Document-level judging of the same corpus scored 0.622 and was rejected (finding 9).

Preference judging is majority reading-order (finding 14). Across the preference cells (§6), four judges from four model families — a 20B MoE, a 30B MoE, a 14B dense model, and a 7B dense model — showed raw first-position preference of 0.85–0.88 wherever they parsed at all; judging every pair in both orders and requiring agreement converted 60–95% of pairs to ties, with only the escape rate varying by judge (decisive rates 0.05–0.40). The bias magnitude is invariant across vendor, architecture, and generation: it is a property of the pairwise task format, not of a model. Order-debiased protocols keep preference verdicts honest at the cost of most of the sample.

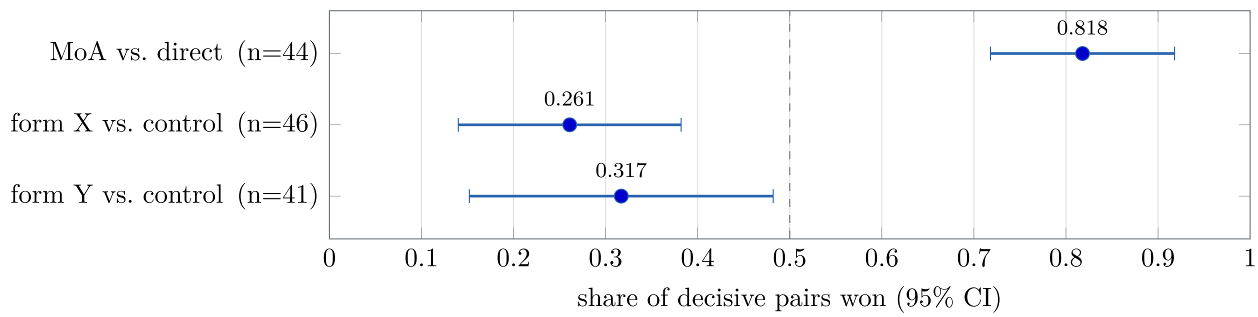

Figure 2: The preference cell (order-debiased; a pair is decisive only when the same side wins both orderings; cluster-bootstrap CIs over cases). The aggregated pipeline beats the matched-length direct baseline decisively; both phrase-instructed arms *lose* to their own control, in counterbalanced forms whose only difference from control is the phrase. The dashed line is indifference.

5 The phrase-swap cell: instructions buy the phrase

Design. Three arms, 18 cases, 7 repeats (378 runs). The control arm is the minimal writer prompt. The two treated arms append a byte-identical clause differing in exactly three words: *"When you state a number that is an estimate rather than an established fact, label it using the phrase 'modeled at' "* (form X) or *"… 'taken to be' "* (form Y). Form X is the historically dictated house form (11 registry entries across 9 prompt symbols); form Y was chosen the day before the run, appears zero times in 2M characters of archive, and matches form X's spontaneous rate exactly (0/249 clean-provenance runs for both). A harness gate refuses to run unless each arm contains exactly its own form and no other registry phrase.

Compliance is form-tracking and the house form has no privilege. Form X appears in 0.667 of its arm's outputs (bootstrap CI vs. control [+0.468, +0.857]); form Y in 0.698 ([+0.524, +0.865]); cross-form leakage is 0/252 — no treated run ever produced the other arm's form, and no control run produced either (Figure 1, left). A phrase drilled through months of system prompts and a day-old synonym perform identically: whatever the instruction engages, it is not a learned disposition toward the trained-on words.

Nothing survives the swap. With each arm's own named phrase subtracted, the validated-judge construct rate is statistically unchanged: form X 0.341 → 0.365 ([−0.079, +0.111]); form Y 0.341 → 0.317 ([−0.103, +0.087]) — opposite nominal directions, both spanning zero (Figure 1, right). Emission is flat across arms (43.7–46.6 sentences), so the null is not bought with silence. The registered minimum detectable effect is +0.224 at the realized cluster count; the result licenses "no form-independent effect ≥ 0.22," not "zero."

The instruction's entire measurable effect is the phrase it names.

6 The preference cell: the lift is real, and the phrase costs preference

Design. Three pairwise comparisons on the same 18 cases (126 pairs each; both orderings; decisive only when the same side wins both; primary judge `gpt-oss:20b`, which also wrote every output on both sides of every pair, making self-preference symmetric by construction). C1 pairs the phrase-swap control arm (the 2-layer MoA) against a generated direct arm: the same writer, same cases, same sampling, with the writer prompt minus only the sentence referencing contributions. C2 and C3 pair each instructed arm against control — and because §5 established those arms differ only in the phrase, any preference gap is causally the phrase's.

The lift replicates at 20B scale. The aggregated side wins 0.818 [0.718, 0.919] of 44 decisive pairs. Arm mean lengths differ by 1.4% (7,757 vs. 7,649 chars), and the share among near-length-matched pairs is 0.815, so the lift is not length (Figure 2). It also does not track domain-framework density (the winner has more frameworks per kilocharacter in only 0.409 of decisive pairs). Its composition is *unidentified* by our feature set — reported as such.

The phrase arms lose, in both forms. The instructed arms' shares against their own control are 0.261 [0.140, 0.388] (form X) and 0.317 [0.152, 0.480] (form Y): both confidence intervals lie entirely below 0.5, in the same direction, across counterbalanced forms, and the penalty survives length matching. Our registered prediction was the opposite (that judges reward the compliance register); the result is a falsification-by-reversal, and is stronger than the prediction: **dictated epistemic marking costs preference points**.

Cross-family replication. A second judge from a disjoint model family (a 30B Qwen3 MoE), selected from three candidates *blind to direction* — the pilot measured only parse and decisive rates, and the selection script never computed which side won — re-judged all 378 pairs. The lift replicates: 0.788 [0.645, 0.917] (n=33 decisive). The form-Y penalty replicates: 0.217 [0.125, 0.333] (n=23). The form-X penalty agrees in point estimate (0.276 vs. 0.261) but its interval spans 0.5 at n=29 decisive and is reported, per the registration, as not replicated at this power — a category distinct from contradiction; no comparison lands on the opposite side. The lift is therefore a two-family result; the phrase penalty holds in three of four judge×form cells.

The tension, stated. Chaining §5 and §6, both links causal: an epistemic instruction installs only a surface form; a preference judge penalizes that form. A pipeline tuned toward judge preference — RLHF-style or best-of-n against a preference model — will therefore shed exactly the epistemic marking that instruction paradigms install, while leaving the underlying disposition (unchanged by the instruction anyway) untouched. Preference optimization and epistemic disposition are not merely unaligned; at this scale they are measurably opposed at the surface where instructions act.

7 Mechanisms the MoA narrative assumes

The aggregator selects sources; it does not recompute. With arithmetic claims planted in proposer texts: when one proposer is corrupted and a clean one remains, the corrupted value reaches the answer 0/27 times; when every proposer is corrupted, 5/27; when the working is also stripped, 12/27. The monotone gradient $1 \to 0 \to 5 \to 12$ across arms (Figure 3) identifies preference for uncorrupted sources with no independent recomputation. MoA's own evidence that the aggregator's output overlaps most with the best proposals [1] is the same phenomenon viewed from the benign side; the planted-error view adds its failure boundary, and shows the "collaborativeness even from worse inputs" claim fails at the fact level: worse inputs made the writer factually worse.

Corroboration is ignored. Claims advanced by $k \geq 2$ independent proposers transport to the final answer no more often than single-proposer claims (0.487 vs. 0.524; difference CI [-0.153, $+0.080$], excluding gains above $+0.08$) — the one reliability signal the architecture uniquely provides is unused.

Transport is weak; invention is real. On de-scaffolded arms under the validated instru-

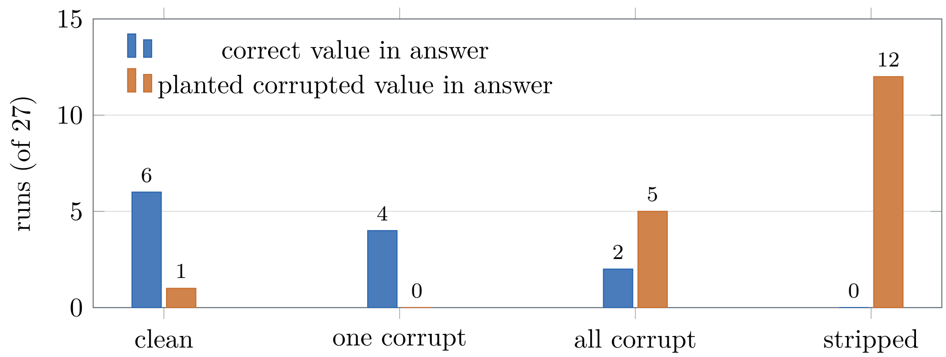

Figure 3: The source-selector gradient (planted arithmetic errors; exact match on known values). As clean sources are removed — one proposer corrupted, all corrupted, working stripped — adoption of the planted wrong value rises monotonically $1 \to 0 \to 5 \to 12$ while correct values vanish. The aggregator prefers uncorrupted sources when they exist and has no independent hold on the content when they do not.

ment the writer transports only a small fraction of the epistemic qualification it is given, and the relation is linear over the observed range (Figure 4). Within-corpus slopes run 0.105–0.326 across two writers and two corpora, three of four excluding zero, with phi4's extension-corpus fit 0.170 [+0.067, +0.266] replicating on nine cases the original corpus never contained. The observed slope is an attenuated *lower* bound: supply is judge-measured with inter-judge $r = 0.528$, so correcting for that attenuation on the original corpus moves 0.150 to 0.217. Pooling the two corpora is declined — their supply regimes differ by a factor of five and the pooled slope reduces to the line between the two clusters. The writer also invents qualification at a 0.333 [0.138, 0.609] rate on inputs confirmed to contain none, an independent estimate that falls inside the intercept's interval.

Mixing is not automatically diverse. Two medical fine-tunes of the *same* base model do not share a domain signature: one shifts in-domain framework density +0.232 [+0.072, +0.412] over the base, the other +0.109 [−0.056, +0.300], and a second lineage shows −0.041 [−0.099, +0.025] — per-model, not per-domain-training. A post-hoc descriptive check (not a registered result) further found two same-domain seats no more distant from each other than one seat is from itself across samples; registered or not, the published Self-MoA result [2] independently reaches the same conclusion at benchmark scale.

Accuracy. On a 60-item self-verifying battery the aggregated pipeline and the bare writer are statistically indistinguishable at a 0.92–0.97 ceiling. Two registered follow-up batteries then attacked the ceiling itself — 36 multi-step computation items and 30 rule-interaction items with script-computed answers, each under a baseline-only difficulty screen — and the direct writer ceilinged again at 0.977 and 0.983. Under the pre-registered two-strikes rule the question is resolved as *untestable for lack of headroom* on well-specified items at this scale: a council cannot add accuracy where the single writer already delivers it.

8 The audit's consequences, engineered and re-tested

A falsification program earns a second obligation: if the findings are real, the design rules they imply should survive prospective tests of their own. We distilled the audit into a runtime

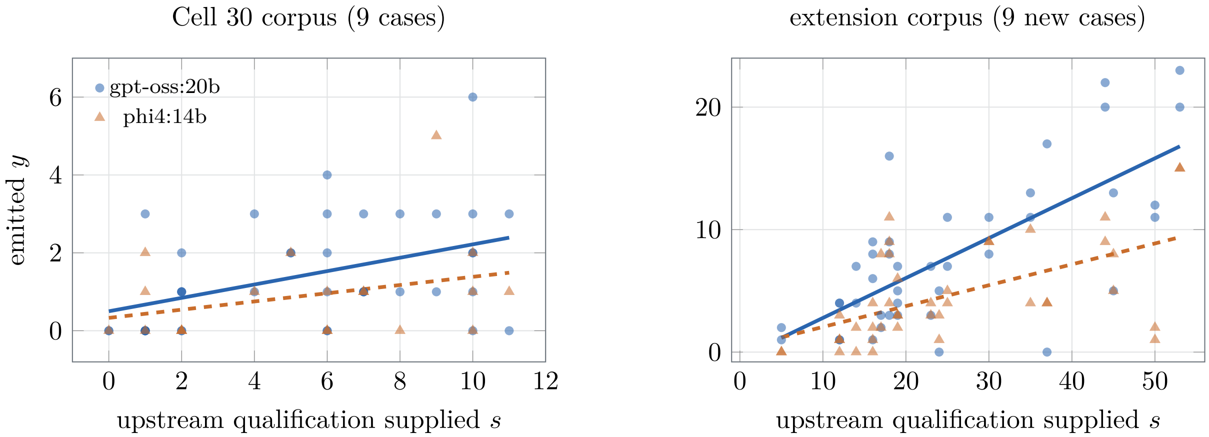

Figure 4: The transport coefficient $y = ws + c$, 144 runs over 18 cases and two writers. Each point is one run: upstream epistemic qualification supplied by the seats against qualification emitted by the writer, both counted by the validated two-judge sentence protocol. Lines are within-panel least-squares fits. **The panels are separated because the two corpora occupy different supply regimes** (mean s 5.4 against 25.3); pooling them produces a slope that is essentially the line between the two clusters, so no pooled fit is drawn. Within-panel slopes: gpt-oss 0.172 [+0.052, +0.297] and 0.326 [+0.207, +0.434]; phi4 0.105 [−0.031, +0.273] and 0.170 [+0.067, +0.265]. Three of the four exclude zero, and phi4's right-hand fit is an independent replication on nine cases the original corpus never contained. The writer emits a small fraction of what it is given, and the fraction is a property of the writing step rather than of one model.

architecture (an evidence-bound harness, released with the ledger) and then registered a cell against each load-bearing component. Three results follow; each is a single-factor causal design, and two of the five verdicts in this arc went *against* our own design rationale.

Engineered redundancy is causal. The source-selector finding (§7) implies protection is coverage, and coverage is a design variable. A registered cell assigned a load-bearing quantity to a second proposer as one factor: corrupted-value adoption fell from 0.900 (sole-sourced, corrupted — the sharpest no-verification number in the program) to 0.450 ([+0.200, +0.725]), and the *clean* value was adopted (0.000 → 0.500, [+0.175, +0.825]) — selection, not dilution. Protection is bimodal across items; two candidate mechanisms for the boundary were then registered and killed in turn (prior-plausibility at 0.544; a stable, near-unanimous elicited ownership map whose prospective prediction landed at exactly 0.500 against a descriptive 0.895 — the confirmation cell doing precisely the anti-calcification work it exists for). The boundary stays open, and the planner's guarantee is stated as expected-value only.

Out-of-band freight carries, and is used. The transport and penalty results (§7, §6) imply epistemic qualification should never route through the writer. The harness therefore carries proposer caveats in a mechanically assembled appendix the writer never sees. Two registered cells tested the channel's halves (Figure 5). Carriage: the appendix delivers 1.000 of both-judge caveat sentences against the 0.174 that survives prose transport, with the preference cost not evaluable at power (0.510 [0.267, 0.754], 65/116 ties — no evidence of a penalty, no license yet to claim its absence). Uptake: a decision-relevant caveat delivered appendix-only flips a reader model's registered two-token decision in 0.727 of reads against 0.036 for the iden-

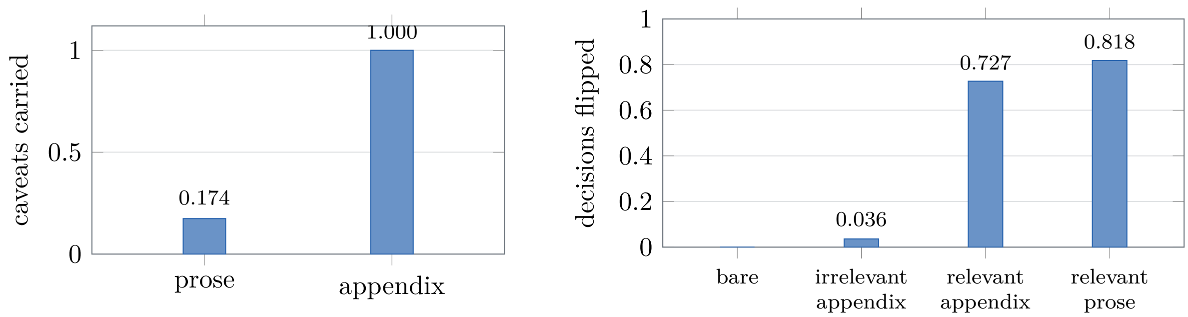

Figure 5: The out-of-band freight channel, both halves. *Left*: share of both-judge caveat sentences reaching the final artifact through writer prose vs. a mechanically assembled appendix (zero-generation design; byte-identical prose in both arms). *Right*: share of registered two-token reader decisions flipped by a planted decision-relevant caveat, by delivery arm (11 items, 440 reads; the irrelevant-appendix arm carries a matched caveat that does not bear on the decision). Uptake is content-specific (+0.691 [+0.455, +0.891] over the irrelevant control) and at parity with in-prose placement (+0.091 [−0.036, +0.218]).

tical appendix block carrying an irrelevant caveat — content-specific, not caution-priming, with direction-balanced items (blocking *and* enabling caveats) so that generic caution cannot masquerade as uptake — and at parity with weaving the same sentence into prose. The channel that preference pressure cannot see is not an exhibit: what it carries changes decisions at no measured transport or preference cost.

An informative null demoted our own isolation rule. The harness isolated proposers (task plus a one-line roster, never sibling outputs) on the strength of a lane-bleed incident from an earlier pipeline version. The registered one-factor test — byte-fixed sibling contributions made visible, production prompts otherwise unchanged — found nothing: off-domain framework density moved −0.009 [−0.048, +0.028] against a pre-registered minimum detectable effect of +0.05–0.08 computed from the archive, with in-domain density unchanged. The isolation rule survives on cost grounds alone (∼10k input characters and 14% longer outputs per proposer, buying no measured change in lane discipline), and its contamination rationale is withdrawn. We report this with the same prominence as the positive results: the architecture documents what its evidence *failed* to support, which is what distinguishes an evidence-bound harness from a prompt stack.

9 What does work: routed re-consultation

Everything above bounds or removes. This section reports the one intervention in the whole program that survived its own registered test, and the structural reason it is not yet worth much.

The design. The aggregator writes its tension list first (our synthesis scaffold mandates one, and 391 of 396 archived runs contain it), and an *orchestrator* — not the aggregator's disposition — parses that list and dispatches a follow-up to the implicated proposer. Three arms over six planted tensions whose deciding fact is withheld from round one: no clarification; a length-matched clarification that is responsive in style but contains *no* deciding fact; and the same clarification containing it. The middle arm exists because §5 makes the alternative live — consultation could shift behaviour as ceremony rather than content.

The result. On runs where the tension was named, the aggregator adopted the fact-implied resolution in 7/7 informed runs against 0.333 in control (cluster-bootstrap difference CI [+0.200, +1.000]) and 0.200 under the contentless clarification ([+0.545, +0.947]). **The information does the work, not the ritual.** The informed arm was unanimous across both judges; resolving every judge disagreement in the other arms adversarially against the prediction still leaves gaps of +0.40 and +0.53; emission is flat across arms.

This is coherent with every failure above rather than an exception to them. Instructions install phrases (§5); corroboration is ignored (§7); the aggregator cannot verify (§7). Routed re-consultation is the only intervention we tested that operates *through* the one capability the aggregator demonstrably has — preference among overlapping sources — by manufacturing an uncorrupted source exactly at the contested quantity.

Why it is not yet worth much: the trigger is scarce. A companion cell asked whether proposers can initiate the loop themselves, giving each a roster and a structured deferral token, then putting the identical sub-question to the proposer whose domain holds the deciding consideration and to one whose domain does not. Routing, when it fires, is accurate: the named domain is the correct one in 10/11 cases, 0.909 [0.623, 0.984] against a two-alternative chance rate of 0.5. But discrimination fails: flag rates are 0.306 out-of-domain against 0.083 in-domain, a gap whose cluster-bootstrap CI [−0.056, +0.528] spans zero, and the in-domain flags are a proposer failing to recognise its *own* competence.

The two cells share one shape:

	mechanism, when triggered	trigger rate
aggregator uses a clarification	7/7	0.34 names the tension
proposer routes a deferral	0.909 correct	0.306 recognises the need
briefed proposer conveys, live	21/21 conveyed, 0.905 adopted	0.208–0.583 by batch

Re-consultation here is recognition-limited, not capability-limited. That is why the orchestrator-routed design is the one we recommend and the proposer-initiated design is not: it draws its trigger from a mandated artefact the aggregator already produces, rather than from a model's assessment of its own competence — the one faculty the program never found evidence for.

The loop closes end-to-end with a live proposer. The results above used controlled clarification texts; two further registered cells replaced them with a real proposer. The first buried the deciding fact in the proposer's own round-one contribution and *halted at its registered pilot gate*: with the fact visible in the pile the tension was named in only 0.208 of runs — recorded as the halt, with the suppression mechanism left unproven. The second restored the original round-one bytes and supplied the fact only as the proposer's private working notes at reply time — knowledge held but never written. The live proposer conveyed the fact in 21/21 dispatched replies, and the aggregator adopted the fact-implied resolution in 0.905 of tension-named runs against the archived control's 0.333 (+0.571 [+0.139, +0.912]), within −0.095 [−0.227, 0.000] of the scripted ceiling. Production, conveyance, and use all work with no controlled inputs; what varies across batches is only the trigger (0.208–0.583 across the three cells that measured it — no single rate should be quoted). One risk rode along: 4/21 live replies embellished the correctly conveyed fact with *fabricated* legal authority (invented case law, invented regulatory rulings), and the aggregator adopted anyway. A content gate screens absent content, not invented content; a registered follow-up cell bounded the rate at 0.100–0.150 reply-level, concentrated on a single trigger topic and name-stable across independent samples, with the gap never self-admitted (0/20 sampled replies).

Scope. Six items throughout, one writer family, and a conditioned n of seven in the original decisive cell, where unanimity was the only way to clear an interval; the live cell's comparators are the original cell's archived arms under a registered, never-pooled fresh-batch check. The judge agreement bar was met on the analysis population (0.703) and not overall (0.583); both are reported. These are existence results about a mechanism, not effect sizes for deployment.

10 Discussion

What the lift is, and is not. The audit does not debunk MoA's headline: aggregation genuinely wins preference at our scale, under a debiased protocol, against a matched-length direct baseline. What it removes is the assumed mechanism story. The lift is not built from the epistemic behaviors the aggregate prompt requests (those requests install phrases, and the judge penalizes them), not from length, and not from domain-framework density; and the aggregator beneath it selects rather than verifies. The residual — plausibly organization, coverage, or specificity harvested from proposers — is the right target for the next instrument, and we decline to name it without one. What the audit does supply constructively is §9: the gain available from an aggregation pipeline is not in instructing the aggregator to be careful, but in routing a specific question back to a specific proposer at the moment a specific claim is contested.

Implication for preference-trained pipelines. The measured tension predicts that preference optimization silently strips epistemic marking. This is consistent with, and sharpens, verbosity- and sycophancy-style results [4, 5]: the pressure is not merely toward longer or more agreeable text but away from the specific surface forms by which pipelines currently signal uncertainty. Systems that must retain calibrated marking under preference pressure will need it carried somewhere preference cannot see — structured fields, tool outputs — rather than in-band prose. We no longer offer that as advice: the out-of-band channel is tested (§8), carries everything prose loses, changes downstream decisions, and shows no detected preference cost.

Implication for evaluation. At auditable scale, five sixths of a raw preference verdict is reading order. Single-ordering LLM-judge evaluations at this scale are measuring position; debiased protocols are usable but must budget for a $\sim$0.35 decisive rate.

11 Limitations

The primary preference judge is also the writer (symmetric within pairs, but family-specific); the cross-family replication discharges this for the lift and for the form-Y penalty, while the form-X penalty rests on one family's interval plus a point-estimate agreement at lower power. One aggregator family and 7–20B scale throughout — the tension should be retested with larger judges. The case battery is advisory-domain and English-only. The phrase-swap cell's null licenses only effects $\geq$0.22; the preference cells' decisive counts (44 and 33) give minimum detectable shares of $\sim$0.67–0.70, which the observed lifts clear but subtler ones would not. Several early program values remain PROVISIONAL pending re-measurement and are not used here. The engineered-consequence cells (§8, §9) carry their own bounds: cluster ceilings of 6–18 items; the uptake verdict rests on 11 of 18 authored items after a registered floor gate; the live-loop cell compares against archived arms under a fresh-batch check rather than concurrent regeneration; the redundancy boundary is bimodal with its mechanism open after two registered attempts; the isolation null covers sibling *outputs*, not cross-domain instruction framing; and the live-reply fabrication rate (4/21) is observed, not bounded. Trigger rates vary by batch (0.208–0.583) and are never optimization targets under the program's Goodhart charter. Claims are stated with their judge-family scope and should not be generalized past it.

12 Reproducibility

The measurement kit (`gst`: record contracts, pure-stdlib statistics, instruments, registry and gates), the frozen dictation registry and form-freeze files, all pre-registrations with their falsification conditions and attainability computations, every deviation and verdict, and the three program audits are in the project repository; each cell's registration is committed before its first run, so temporal ordering is verifiable from history.

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